

Prince Onukwili

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Current role	DevOps Engineer Platform Engineer Cloud Engineer at Professional Development - Career Break
Experience	DevOps, Blockchain, SaaS, Education, Fintech, Social Media
Technologies	Kubernetes, GCP, Bash, Terraform, Prometheus, Grafana, loki, DevOps, Kafka, Redis, AWS, Jenkins, DigitalOcean, SQL, Azure...

I like to work with people who...
Are communicative, available, and easy to collabotate with

Work experience

DevOps Engineer | Platform Engineer | Cloud Engineer,
Professional Development - Career Break 🏠

Mar 2026 - Present (4m)

Professional development and upskilling
DevOps

- Kubernetes
- GCP
- Bash
- Terraform
- Prometheus
- Grafana
- loki
- DevOps

- Architected a PostgreSQL cluster and microservices on Google Kubernetes Engine (GKE) that **sustained 160k+ requests/hour (~3.8M+/day) and 10k+ Postgres transactions/hour at a 99.99% success rate**
- Achieved an 80–100% block rate on a simulated DDoS attack** against the GKE infrastructure using Cloudflare rate-limiting and Google Cloud Firewall rules, while **maintaining a 100% success rate on baseline user traffic**.
- Architected a highly-available PostgreSQL cluster on GKE using the Cloud Native Postgres Operator, **running 3 replicas across different zones with graceful switchover**, daily/weekly GCS backups (disaster recovery), and end-to-end traffic encryption via cert-manager + TLS + Google-managed encrypted disks
- Created Service Level Objectives (SLOs) and Error Budget for critical services (99.5% availability and latency)**. Monitored their SLIs using Prometheus + Istio metrics, and **handled Incident response whenever error or latency rates spiked using PagerDuty**
- Deployed a NetBird VPN with routing peers and Cloud VPC network routes, **giving remote team members private access to internal resources** (GKE, databases, etc.) **with no internal resource exposed to the public internet**
- Implemented a Shared VPC architecture across the Google Cloud organisation, enabling all projects and environments (staging, prod) to share the same internal network without VPC peering.
- Provisioned all infrastructure as code using Terraform and Terragrunt, modularised across environments.
- Enforced end-to-end encryption and mTLS** across all cluster services **using a hybrid Istio Ambient + Sidecar mesh**
- Ambient mode for most workloads to conserve resources, sidecar mode for microservices requiring detailed tracing.
- Managed internal and external Certificate Authorities using Cert Manager.
- Self-managed TLS for internal services and ACME Let's Encrypt + Cloud DNS for external-facing apps.

Manage your digital assets on the cloud
Blockchain

Kubernetes

GCP

Kafka

Redis

Terraform

Prometheus

Grafana

Loki

- **Reduced Cloud Logging costs from \$300/day to <\$1/day** by scoping Google Kubernetes Engine (GKE) logs to audit, system, and event logs only, and forwarding container logs to a self-managed Loki + Grafana stack.
- **Reduced Cloud Network Egress costs from \$2,500+/month to <\$100/month** by co-locating Bitcoin RPC nodes, indexers, and CockroachDB in the same region/zone, eliminating cross-region data transfer.
- **Reduced overall monthly Cloud costs from ~\$11,500 to ~\$7,500 within 2 months** by right-sizing Kubernetes resources and switching Cloud Run to request-based billing.
- Architected a highly-available, fault-tolerant CockroachDB cluster on GKE that **sustained 50k+ blocks/hour (7.5M+ Ethereum transactions/hour) for days on end** while simultaneously serving live user traffic.
- Implemented end-to-end encryption of data in transit using Mutual TLS encryption, auto-renewing TLS certs via cert-manager, daily/weekly GCS snapshots, and Pod Disruption Budgets.
- **Live-migrated Keycloak and APISIX + ETCD from bare GCP Virtual Machines (VMs) to GKE with zero downtime**, including a Keycloak DB migration from MySQL to PostgreSQL with all users, clients, and realms intact.
- **Provisioned Bitcoin and Ethereum RPC nodes on GCP** Blockchain Node Engine and GKE for blockchain data indexing.
- Built CI/CD pipelines for each microservice using Workload Identity Federation (no SA keys), publishing container images to Artifact Registry and deploying to Cloud Run.
- **Deployed Falco for runtime threat detection in GKE with Slack alerting**, and integrated Trivy image scanning into CI/CD pipelines.
- **Enabled encryption of data-at-rest** using Google-managed disk encryption

Enabling teams deploy applications on the Cloud seamlessly
SaaS

AWS

Terraform

Jenkins

- Created “Konstacks” on AWS Cloud provider - pre-configured infrastructure which **enabled devs and founders to easily deploy their applications on AWS infrastructure without needing in-depth experience of AWS services**
- **Provisioned pre-configured AWS ECS Fargate environments, RDS databases, ECR repositories**, etc, enabling devs to easily set up their applications on AWS services
- Provisioned AWS EC2 instances with Autoscaling enabled, fronted with an External Application load-balancer, enabling traffic from the public internet to the EC2 autoscaling group
- Created Network Security Groups, **preventing external access to internal applications, except through the External Application Load Balancers**
- Ensured **least-privileged access to AWS resources** by creating IAM policies with only the required permissions for CI/CD pipelines and Terraform
- Automated the entire infrastructure set-up and destruction using Terraform + Jenkins CI/CD pipeline

An E-learn Platform for students and schools
Education

AWS

DigitalOcean

DevOps

- I supported a growing startup in the EduTech space as they prepared to launch their platform.
- Like many early-stage companies, they needed a reliable infrastructure strategy that would support both development and live environments, which didn't cause them to overspend on unnecessary resources
- **Initial AWS Architecture (Development Phase)**
 - Designed a cost-effective cloud setup using AWS services to support rapid development and testing.
 - Deployed serverless functions on AWS Lambda for components like the notification service.
 - Used AWS ECS Fargate to run the Django-based web server in Docker containers.
 - Configured AWS S3 for storing media and static assets, and AWS ECR for container image storage.
 - Set up a lightweight EC2 instance to run PostgreSQL, Redis, and background workers—providing more control at lower cost than managed alternatives.
- **Full Migration to DigitalOcean (Production Phase)**
 - Led the migration from the AWS development environment to a self-hosted DigitalOcean VM (with high compute capacity) to reduce costs during traffic surges.
 - Set up Coolify, a web-based deployment tool, to manage and run all applications and the reverse proxy setup.
 - I created CI/CD workflows with GitHub Actions, building Docker images, pushing them to GitHub Container Registry, and re-deploying them on the VM.
 - Provisioned PostgreSQL and Redis directly on the VM, with cron jobs for automated daily backups to AWS S3.
 - Finally, I updated the platform's domain settings on Namecheap, so everything pointed to the new server (including support for subdomains).

Fintech

SQL

Terraform

Azure

DevOps

- **[CHALLENGES]**
 - Developers manually published updates to an internal NuGet feed, leading to inconsistencies and error-prone versioning.
 - Each app (microservice) was maintained in its own repository, with no automated way to deploy updates independently.
 - Manually creating resource groups and configuring cloud resources for each app resulted in repetitive work and potential misconfigurations.
- **[SOLUTIONS]**
 - I automated the versioning and publishing of .NET class libraries on Nuget using Azure DevOps CI/CD pipelines
 - I reduced manual errors in the creation of resources on Azure Cloud by automating their creation using Terraform.
 - Creating resources such as Azure Functions Apps, SQL databases, App Insights, Key Vaults, Service Bus, etc...

- I automated and sped up the deployment of each microservice application on Azure Function Apps across staging, and production environments, seeding and migration of the SQL databases using Azure DevOps CI/CD pipelines
- I created various role assignments to authenticate the Function Apps access to other resources within Azure Cloud (e.g KeyVaults, Service Bus namespaces, and redis) using managed Identities.
- **[RESULTS]**
- Developers no longer need to manually publish packages or manage versioning.
- Deployment errors and manual overhead were significantly reduced.
- A consistent, scalable CI/CD workflow was established across repositories, freeing the team to focus on building new features and innovation.

Lead backend developer (freelance) - Node Js, Express Js, Typescript, MongoDB, Cloudinary, Socio Africa (Startup) 🌱

Jul 2024 - Nov 2024 (3m)

A social media platform for sharing ideas and opinions
Social Media

React MongoDB Node.js Next.js DigitalOcean

- Co-developed Socio Africa, an African-led social media platform as a founding software engineer, contributing to its rapid growth to over 10,000+ active users in just months.
- Developed the server-side application with Node.js, Express.js, TypeScript, and MongoDB to create robust API endpoints and databases.
- Designed and applied algorithms for post features, such as creation, editing, commenting, and reactions, promoting user engagement and boosting daily active users by 30%.
- Optimized database query performance through advanced indexing, resulting in a 50% reduction in query latency and a significant enhancement in user satisfaction.

Full stack developer - React Js, Next Js, Node Js, C#.NET, Microsoft SQL, Momas LTD Nigeria 🏢

Jul 2022 - Jul 2023 (1y)

- I streamlined application development by 20% through GOPack, a product I engineered. It simplifies bundling projects into static files using pre-configured webpack, curbing plagiarism and expediting production-ready setups.
- Boosted customer turnout by 15% through a payment web app enabling easy energy/power payments using technologies like PayStack, Remita, and React Js.
- Created a real-time global personnel tracking software using MQTT, React Js, Node Js, and Google Maps, yielding approximately 10% revenue increase for the organization.

Education

Nnamdi Azikiwe University

Oct 2023 - Present (2y 9m)

B.Sc (4.73) Software engineering

Aitech computer school Nnamdi Azikiwe University

Oct 2021 - Apr 2022 (6m)

Certification Certification in Software Engineering

Federal science and technical college, Yaba

Sep 2018 - Sep 2021 (3y)

Software Engineering

Snippets

Simulated DDoS attack against GKE infra; Achieved an 80% - 100% BLOCK RATE ON ATTACKS; While a 100% SUCCESS RATE ON BASELINE; Protected infra using Cloudflare

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Ran a preamble of a DDoS attack on my apps running on GKE, protecting the infrastructure using Cloudflare + GCP Firewall Rules.

- Was able to **achieve an 80% - 100% Block rate** on the attack traffic, and
- A **100% success rate on the baseline** traffic (normal user traffic)

SIMULATED DDoS ATTACK AGAINST GKE INFRA, PROTECTING WITH CLOUDFLARE AND GCP FIREWALL

Achieved an 80% - 100% Block Rate on Attack Traffic
100% Success Rate On Baseline ("normal") Traffic



Built an infrastructure on GKE which withstood 160k+ reqs in 1 hr (~3.8M+ reqs/day); Achieved a 99.99% success rate; Also scaling PostgreSQL cluster to support 180+ Transactions/second

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Architected my Infrastructure on GKE comprising Istio Gateway, a microservice and Postgres Cluster, to **handle 160k+ requests in 1 hour with a 99.99% success rate**

BUILT GKE CLUSTER TO WITHSTAND 160K+ REQ 1 HOUR

ACHIEVED A 99.99% SUCCESS/AVAILABILITY RATE
POSTGRES CLUSTER SUPPORTING UP TO 180+ TRANSACTIONS EVERY SECOND
EQUIVALENT TO ~3.8+ MILLION REQUESTS PER DAY

Architected a PostgreSQL cluster on GKE to WITHSTAND TRAFFIC BURSTS, properly RECOVER FROM DISASTER, ENCRYPT DATA AT REST & IN TRAFFIC, implement CONNECTION POOLING

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Set up a Cloud Native PostgreSQL Cluster on GKE with the ability to withstand Traffic Bursts, Recover from Disaster, Encrypt Data in transit and at rest, etc...

SETUP A POSTGRESCL CLUSTER ON GKE USING CLOUD NATIVE POSTGRES OPERATOR

WITHSTAND TRAFFIC BURSTS
RECOVER FROM DISASTER
ENCRYPT DATA AT REST & IN TRAFFIC
Handle 1000s of simultaneous connections

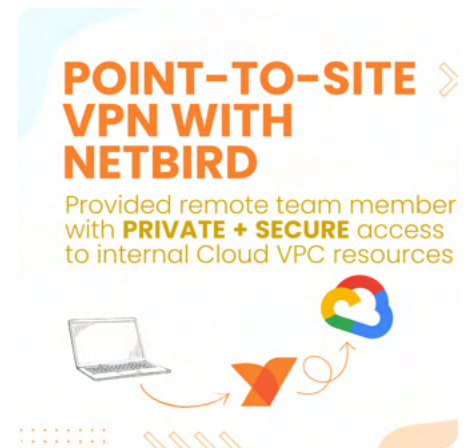


Provided remote team members with PRIVATE + SECURE access to internal Cloud VPC resources — Netbird

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As an org your remote teams are bound to access your internal Cloud resources....but, **how do you provide them with access to your internal Cloud network??**

I was able to implement a **point-to-site VPN**, providing my remote teammates with a **PRIVATE and SECURE access to our Cloud Network**



SAVED \$840+ ANNUALLY on setting up my blockchain app on self-managed Kubernetes Cluster

@ www.linkedin.com

I ran an Ethereum blockchain system on Google Cloud — without breaking the bank.

Instead of using Google's expensive Kubernetes service, I set up a custom version that saved over \$70 every month, while still giving us all the features we needed to stay fast and stable.



More about me

I like to work with people who...

Are communicative, available, and easy to collabotate with

Interests

Writing Articles and tutorials,
Blogging on LinkedIn,
Watching Netflix,
Family and friends time,